

SPRINT PROGRESS REPORT

Post-Stroke Gait IMU Classification

27 August – 4 September 2026

MR-ICT Systematic Review · Wearable IMU Gait Classification

OVERVIEW

Sprint at a Glance

Eighteen leakage-safe experiments re-tested the lower-back-first question, probed enrichment and synthesis, and screened new external data.

18

Experiments executed

fully leakage-safe

1

Model improvement admitted

lower-back ensemble

11+

Rescue / enrichment attempts

rejected on evidence

3

New datasets identified

candidates for acquisition

Every rejection this sprint followed the same predefined non-inferiority gate — the absence of a bigger win is a disciplined negative result, not a stalled project.

What Changed This Sprint



Admitted

- Fixed equal-probability ERM + Deep CORAL + ERM++ ensemble on lower-back acceleration — passed every non-inferiority and material-gain gate
- A formal cross-dataset evidence gate that now governs every future enrichment attempt
- A rigorous test-set credibility audit quantifying exactly how much more paired data is required



Rejected

- Lower-back-only as a replacement prototype — failed the external gate (AUROC 0.757 vs 0.914)
- MAREA / DUO-GAIT / NONAN healthy-pool enrichment — introduced domain-shift false positives
- Six synthetic-data generators — interpolation, recombination, 2× VAE, diffusion
- Signed-axis representation, pooled gyroscope, MiniROCKET fusion, MIL, canonical architecture swaps

Lower-Back ERM + CORAL + ERM++ Ensemble

Protocol

- 314 participants, 22,506 lower-back windows — Felius + Voisard + Sint
- Leave-one-complete-source-out outer evaluation, 5 seeds
- Fixed equal-probability average of ERM, Deep CORAL, ERM++-style head warm-up
- RevalExo and NONAN not loaded for training, tuning, or selection

Metric	This sprint	Prior	Δ (95% CI)
AUROC	0.8882	0.8737	+0.0146 (+0.001 to +0.032)
Balanced accuracy	0.8140	0.7834	+0.0306 (+0.004 to +0.064)
Specificity	0.7743	0.7145	+0.0599 (−0.001 to +0.132)
Brier score	0.1425	0.1664	−0.0239 (−0.049 to −0.002)

Every gain shown is the paired-bootstrap delta (ensemble – plain ERM) at 95% CI — a statistically supported, evidence-gated improvement, not a lucky seed.

Do Our Methods Actually Match the Literature?

A targeted audit checked whether prior “failed” methods matched the published implementations, or only reduced local approximations of them.

Previous attempt	Audit verdict
Current CNN	Valid compact CNN — but not canonical InceptionTime
CORAL	Core implementation closely matches HAROOD; credible for the tested weight
ERM++	Only an ERM++-style subset, not the complete ERM++ recipe
GroupDRO	Used source×class groups instead of source-domain groups — family-level rejection was invalid
MiniROCKET	Used 2,000 features + fixed heads; canonical 10,000-feature scaled RidgeCV was untested
Attention MIL	Attention equation is faithful, but gait bags are a project-specific adaptation
Healthy DDPM	Not IMUDiffusion — architecture, STFT, epochs and objective differ materially

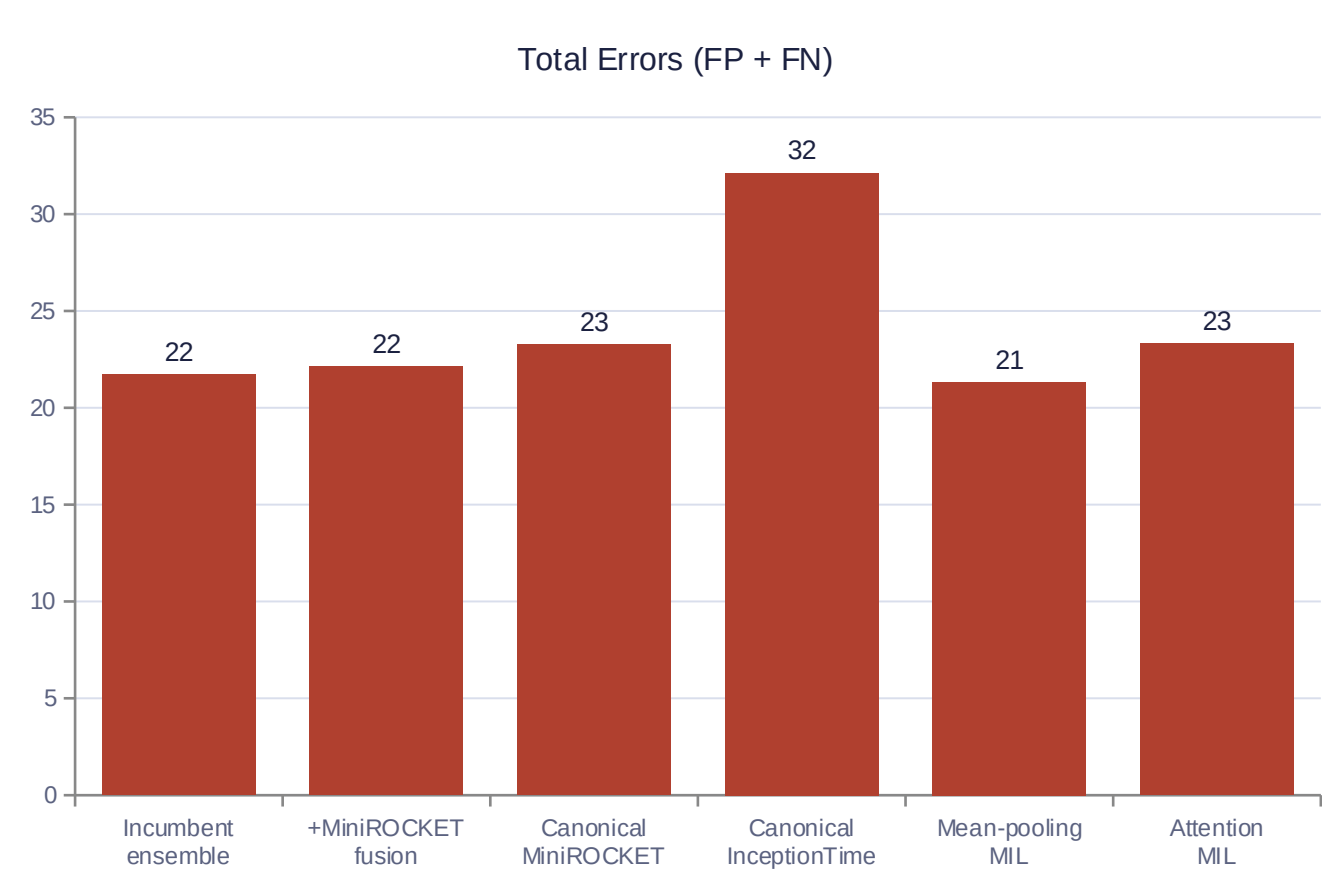
Correct name for the current model

Compact Inception-style ERM + HAROOD-style CORAL + ERM++-style ensemble

It must not be presented as canonical InceptionTime or complete ERM++ — the audit is a naming and credibility correction, not a new rejection.

Rescue Attempts After the Ensemble

Every candidate had to beat the incumbent (21.73 total errors per source/seed) under the same three-source, five-seed, leakage-safe protocol.



Candidate	AUROC	FP	FN	Verdict
Incumbent ensemble	0.8882	9.80	11.93	Selected
+ MiniROCKET fusion	0.8869	9.33	12.80	Rejected
Canonical MiniROCKET	0.8572	9.40	13.87	Rejected
Canonical InceptionTime	0.8273	13.07	19.07	Rejected
Mean-pooling MIL	0.8442	9.80	11.53	Rejected
Attention MIL	0.8043	11.00	12.33	Rejected

Why More Data Is the Limiting Factor

Seed-Consensus Confusion (development)

	Pred: Healthy	Pred: Stroke
True: Healthy	TN: 99	FP: 27
True: Stroke	FN: 33	TP: 155

Pooled specificity 78.6% (CI 70.6–84.8%) · sensitivity 82.4% (CI 76.4–87.2%). Diagnostic summary, not an independent test.

Planning target

~257

independent healthy participants needed

~222

independent stroke participants needed

Target: a two-sided 95% Wilson interval no wider than 10 points at current development rates — illustrating why 7 healthy / 10 stroke (RevalExo) cannot be a reliability claim.

New Datasets This Sprint

Already tested (reporting-only)

Carpinella 6MWT healthy cohort

60 healthy participants, 6,109 windows — lower-back-only baseline made 0 stroke predictions at 0.50 (Wilson upper bound 6.0%)

Zenodo stroke-only cohort

3-channel detected 9/10 stroke (95% CI 59.6–98.2%); lower-back-only detected 10/10 (95% CI 72.2–100%)

Triaxial older-healthy cohort

53/59 (89.8%) false positives — diagnosed as a signal-domain/device mismatch (0.578g vs the model's fitted 1.016g), not an age effect

Screened for future acquisition

Source	Composition	Role
WearGait-PD	85 healthy + 100 Parkinson's, full-body IMU	Non-stroke hard-negative test
6MWT normative	60 healthy, lower-back only	Healthy age/domain reference
BLISS	21 mixed impairment	Foot/impairment stress test

None of these three are acquired or loaded yet — WearGait-PD is the priority pull, since it directly targets the 52.2% non-stroke false-positive rate.

Cross-Dataset Evidence Gate

Eight literature/code bases were audited against this project's exact task, sensors, and split rules before any future modelling approach is proposed.

Why past negatives don't contradict the literature

- Many reported successes concern generic HAR or gait detection, not disease classification
- Many use target-domain data during fitting — this project explicitly forbids that
- Accelerometer-only can outperform accelerometer + gyroscope — consistent with this project's own finding
- Synthetic gains can reverse for individual held-out people or larger configurations

Mandatory checklist for every future experiment

- Official-code trace before re-implementing any method
- Matched ERM baseline under the same protocol
- Participant/source-disjoint validation, always
- Training-only generator fitting; real-only validation/testing
- Worst-source clinical metrics reported, not just pooled means
- No RevalExo/NONAN used for model selection — ever

Where This Goes Next

①

Freeze & Package

Lock the exact lower-back ensemble recipe — no further tuning on these 314 participants. (In progress: notebook 35)

②

One Frozen External Pass

Evaluate the packaged ensemble on RevalExo exactly once, under a pre-registered protocol, without retuning.

③

Recruit New Paired Cohort

Target ~257 healthy / ~222 stroke participants — this sprint's identified limiting requirement.

④

Non-Stroke Specificity

Acquire WearGait-PD or equivalent to test the 52.2% non-stroke false-positive rate against a genuine external cohort.

Closed for now

- Architecture rotation on the existing 314 participants (canonical benchmark closed this)
- Healthy-pool enrichment without source-conditional calibration
- Synthetic data generation without a materially better realism gate

Guardrails carried forward

- RevalExo and NONAN stay frozen — never used for model selection
- Every future experiment must pass the cross-dataset evidence-gate checklist
- No claim of clinical readiness from development-only evidence